

Class - 12

Probability Rules

Probability

ADDITION RULE

The addition rule tells us the probability that **either event A or event B (or both)** will happen.

Basically in addition rule of probability we add the probability of both the individual events and subtract the probability of intersection part so it cannot change the probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

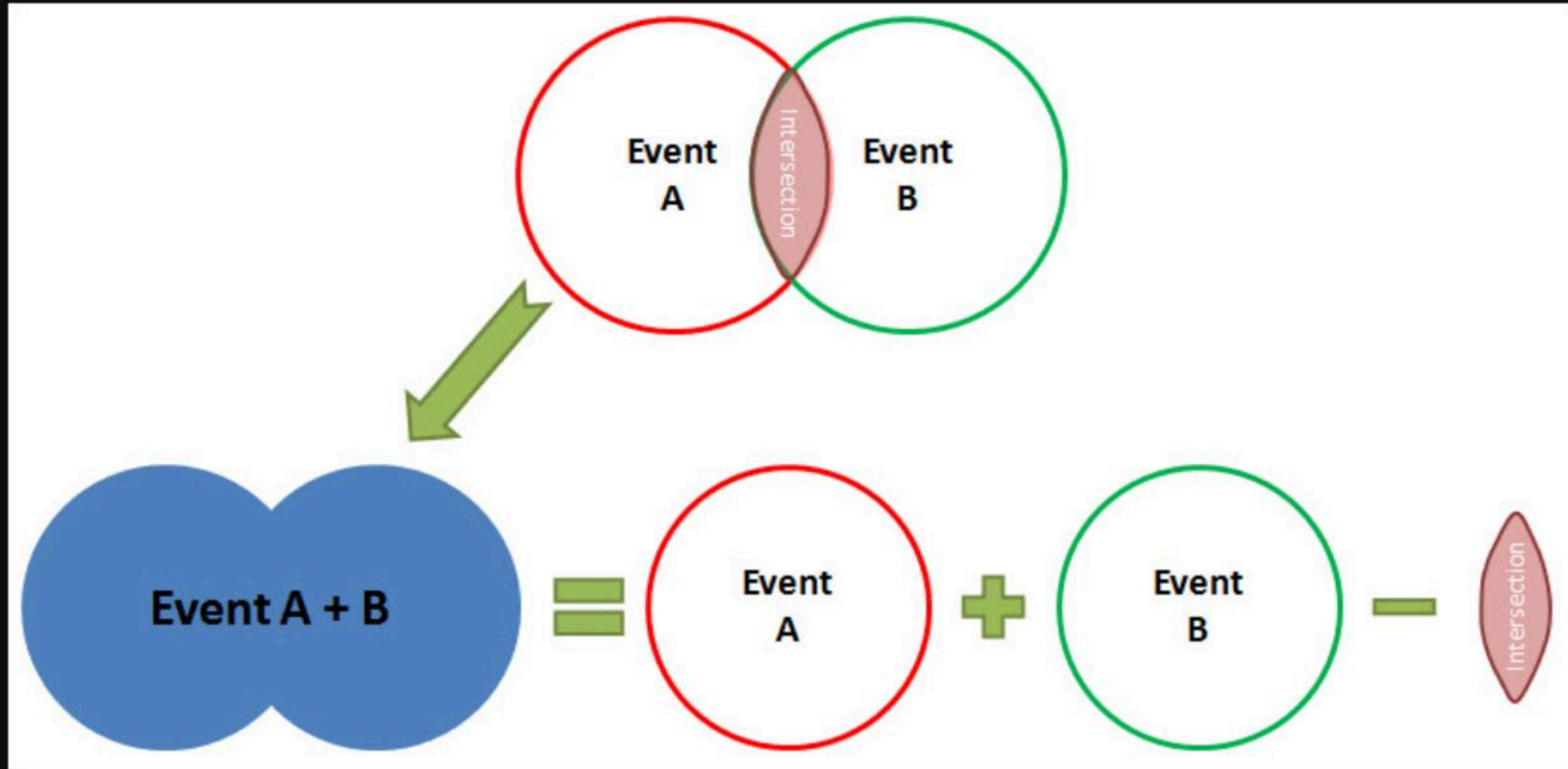
ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- $P(A \cup B)$: Probability of occurring **A** and **B**
- $P(A)$: Probability of **Event A**
- $P(B)$: Probability of **Event B**
- $P(A \cap B)$: **Intersection** of event **A** and **B**

We **subtract** the **intersection** part so it does **count double**.

ADDITION RULE



Venn Diagram

ADDITION RULE

🎲 Example: the probability of drawing a club or a king card from a card deck

Event A - Getting a club.

Event B - Getting a king.

ADDITION RULE

Special Case : Mutually Exclusive event

Now here the addition rule change for the mutually exclusive event because there is nothing in common in them so that the rule is

$$P(A \cup B) = P(A) + P(B)$$

ADDITION RULE

Special Case : Mutually Exclusive event

Getting a King or a Queen from the deck of Cards.

MULTIPLICATION RULE

The multiplication rule is used to find the probability that **two events happen together** (A and B).

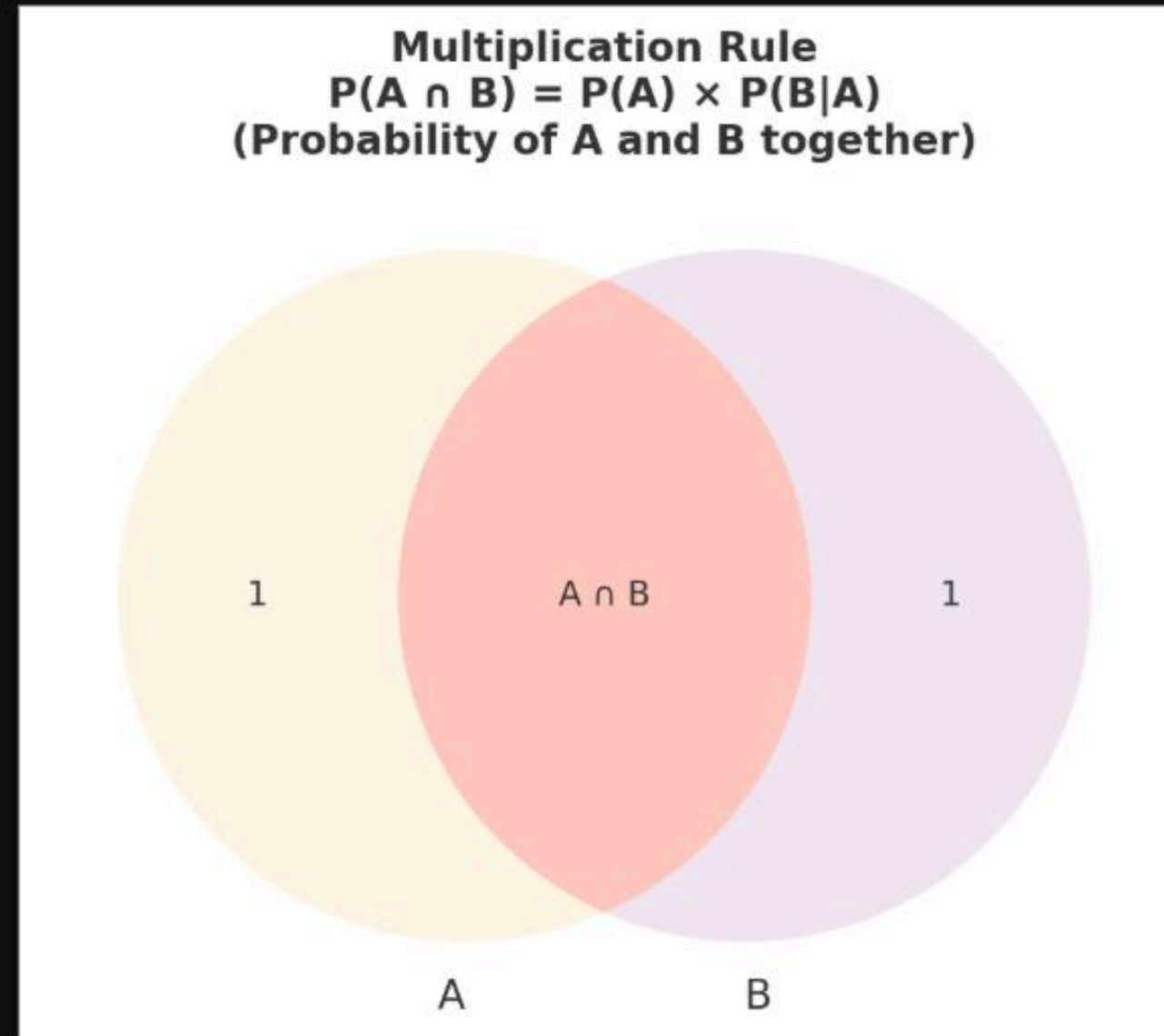
Multiplication rule of probability is used to calculate the probability of two events **occur together**.

- **General rule**

$$P(A \cap B) = P(A) \times P(B|A)$$

- $P(A)$ = probability that event A happens.
- $P(B|A)$ = probability that event B happens **given A already happened**.

MULTIPLICATION RULE



Venn Diagram

MULTIPLICATION RULE

🍫 Example: Picking Chocolates from a Box

A box has **5 milk chocolates** and **3 dark chocolates** (total = 8).

- **Event A:** First pick is **milk**
- **Event B:** Second pick is **milk again** (without replacement)

- $P(A) = \frac{5}{8}$

- $P(B|A) = \frac{4}{7}$

$$P(A \cap B) = P(A) \times P(B|A)$$

$$= \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$$

MULTIPLICATION RULE

Special Case: Independent Event

In the independent event both are independent and does have no effect of happening another event at the same time so the formula of multiplication changes to :

$$P(A \cap B) = P(A) \times P(B)$$

MULTIPLICATION RULE

🎯 Example: Coin Toss + Spinner

- **Event A:** Toss a coin → Get **Heads**
- **Event B:** Spin a spinner with 4 equal sections → Land on **Red**
- $P(A) = \frac{1}{2}$
- $P(B) = \frac{1}{4}$
- $P(A \cap B) = P(A) \times P(B) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

There is no multiplication rule for mutually exclusive events because they can't happen together

CONDITIONAL PROBABILITY

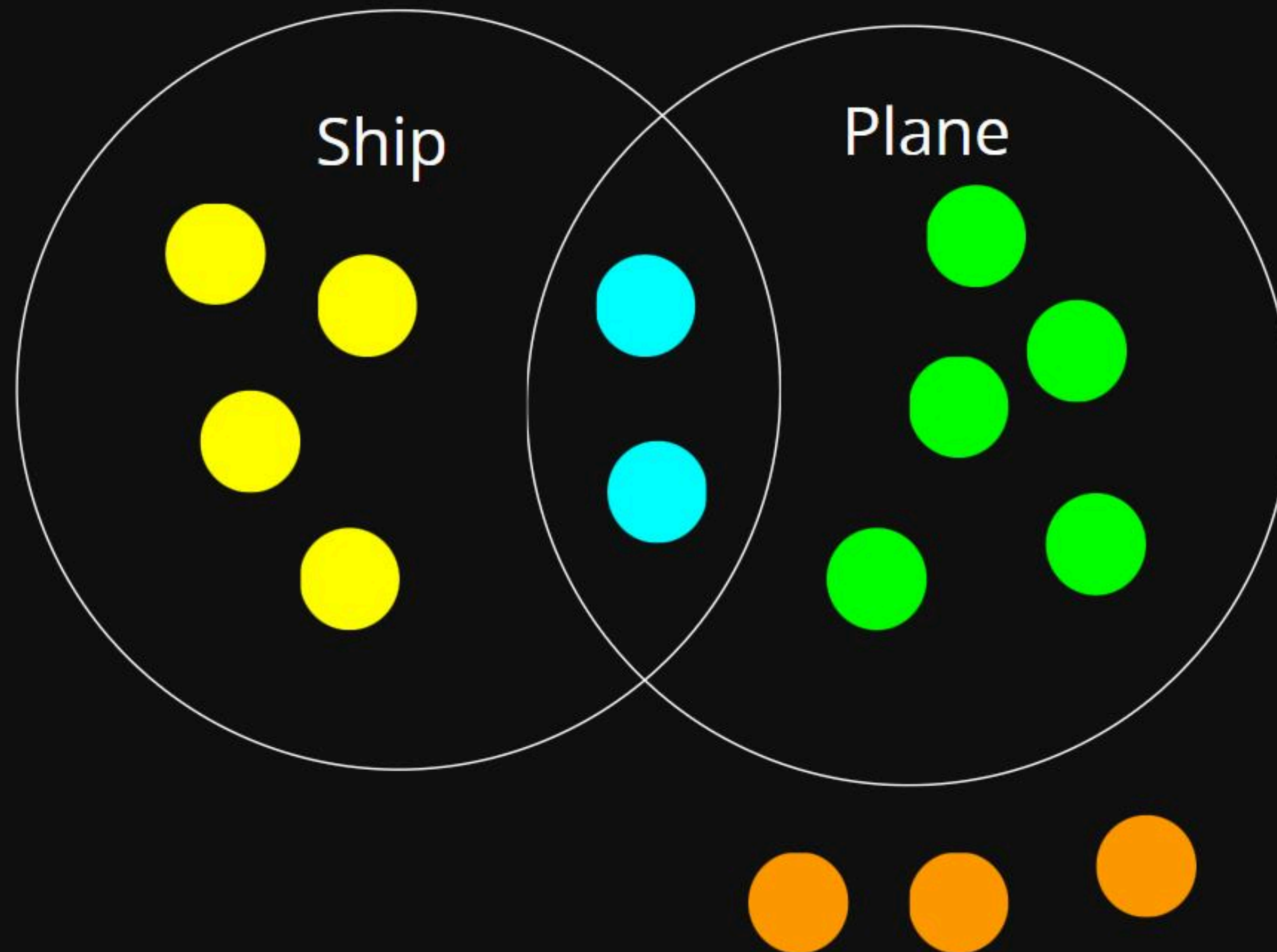
Conditional probability is the probability that an event **B happens given that event A has already happened.**

In **conditional probability** there are two probabilities in which one is already happened.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0$$

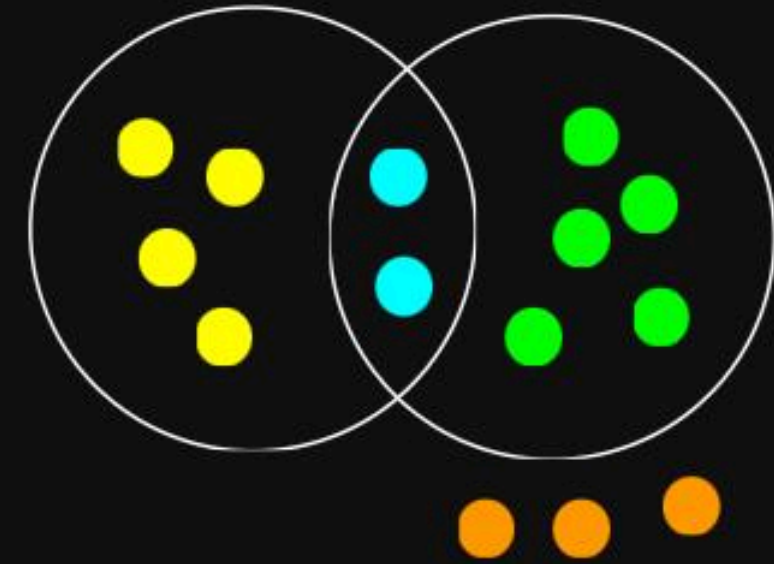
- $P(B|A)$ = probability of B **given A**.
- $P(A \cap B)$ = probability of both A and B happening.
- $P(A)$ = probability of A.

CONDITIONAL PROBABILITY



CONDITIONAL PROBABILITY

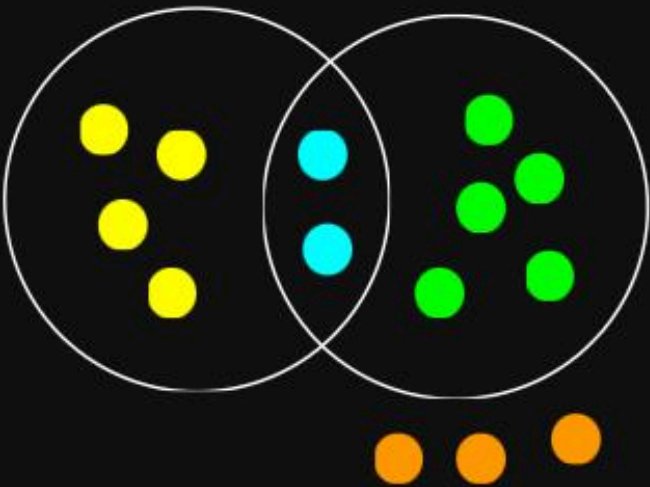
Contingency table



	Loves ship	no ship
Loves plane	2	5
No plane	4	3

CONDITIONAL PROBABILITY

Contingency table

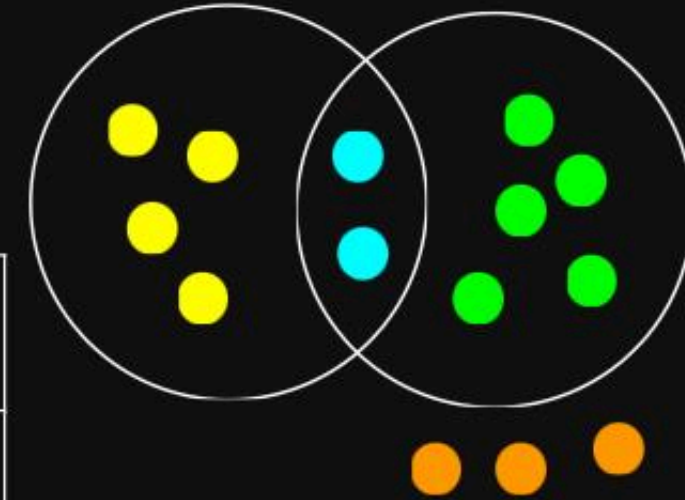


	Loves ship	no ship	row total
Loves plane	2 p - 2/14	5 p - 5/14	7 p - 7/14
No plane	4 p - 4/14	3 p - 3/14	7 p - 7/14
col total	6 p - 6/14	8 p - 8/14	

CONDITIONAL PROBABILITY

Contingency table

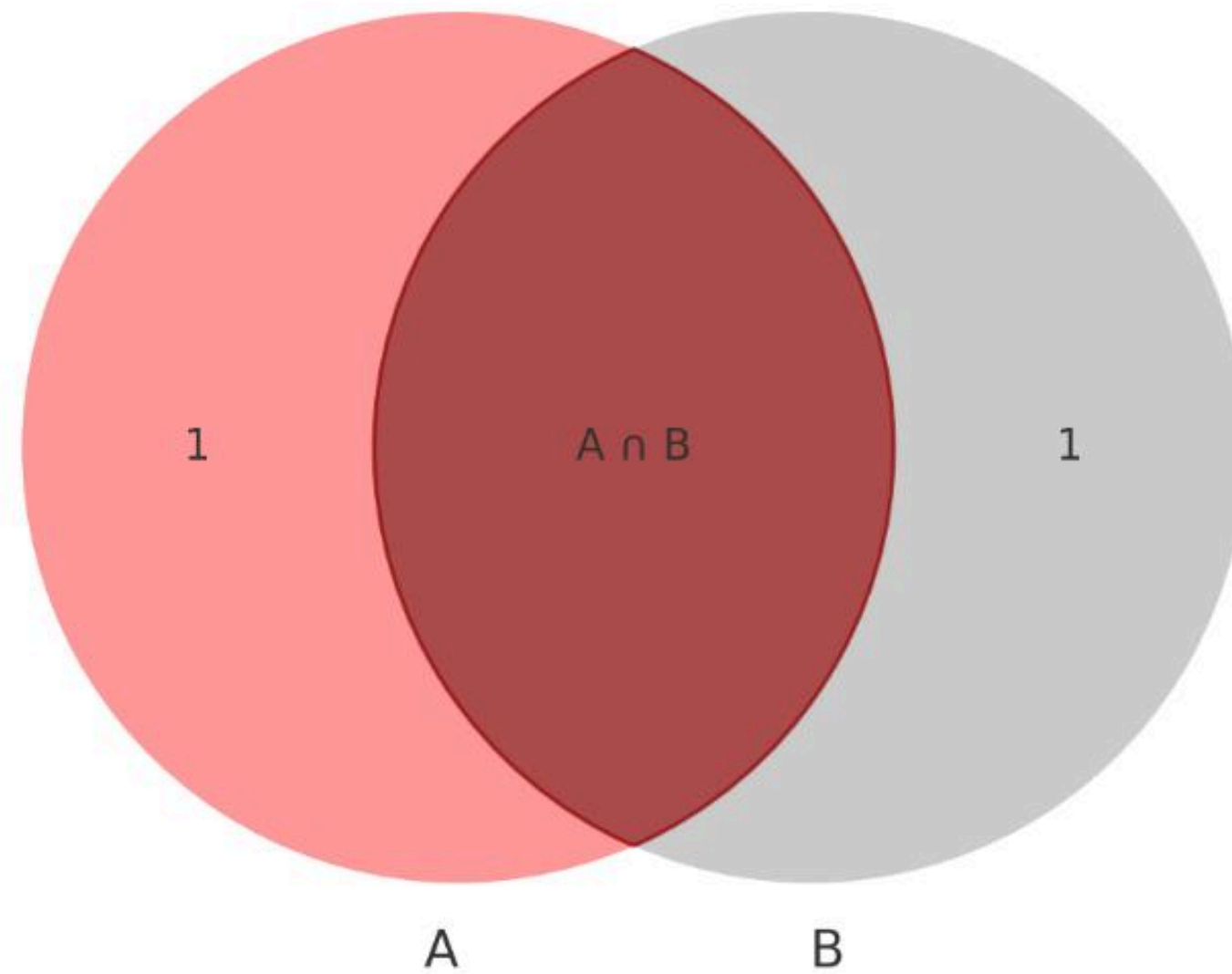
	Loves ship	no ship	row total
Loves plane	2 p - 2/14	5 p - 5/14	7 p - 7/14
No plane	4 p - 4/14	3 p - 3/14	7 p - 7/14
col total	6 p - 6/14	8 p - 8/14	



So this is what conditional probability is ... $P(B|A) = \frac{P(A \cap B)}{P(A)}$

CONDITIONAL PROBABILITY

Conditional Probability
 $P(A|B) = P(A \cap B) / P(B)$
(Probability of A given B already happened)



Venn Diagram

CONDITIONAL PROBABILITY

Easy Example 🎴 :

- Deck of 52 cards.
- Event A = card is a **face card** (J, Q, K $\rightarrow$ 12/52).
- Event B = card is a **King**.

$$P(B|A) = \frac{P(\text{King and Face Card})}{P(\text{Face Card})}$$

$$P(B|A) = \frac{4/52}{12/52} = \frac{4}{12} = \frac{1}{3}$$